

respond to attacks without human interference.

Incident response system:

AI and ML based systems help in providing incident responses, enabling organisations to manage security alerts appropriately. AI automated incident responses mitigate vulnerabilities and deliver faster responses to such events.

Fraud detection: AI and ML based systems can be used to create models to recognise fraud-related patterns. As more data is fed to the system, the AI model becomes more accurate.

Asset discovery: AI and ML can be used for automating the discovery of all key devices and applications. This can play a huge role in mitigating risks.

Key open source AI and ML cyber security tools

Open source cyber security tools help organisations to protect their devices, data, and user landscapes from internal and external threats. These tools can be proactive or reactive, allowing organisations to test systems and check for vulnerabilities or monitor active systems to pre-empt incoming attacks.

Cyber security tools have the following important features:

- Business-need alignment and organisation readiness
 - Highly scalable
 - Support heterogeneous environments and can adapt easily
 - Have proper industry support for technology
 - Integrate easily with an organisation's systems and tools
- Table 1 highlights the most popular

and important open source AI and ML based cyber security tools.

Benefits of AI and ML in cyber security

Integrating AI and ML into a cyber security system has quite a few benefits:

- Ability to detect nuanced attacks, strengthen security, and enhance incident response
- Improves the detection and response cycle time
- Organisations can rapidly quantify risks and accelerate analyst decision-making with data-driven mitigation measures
- Prevents and mitigates cyber security breaches and malicious attacks
- Improved workforce experience
- Improved customer satisfaction and brand reputation due to heightened cyber security protection and increased trust in the organisation's security protocols

Cyber threats have become innovative and are constantly evolving. Also, data is filled with new patterns that are hard to capture and analyse manually. AI and ML offer a powerful way of identifying vulnerabilities,

threats and attacks across organisations as well as social media. AI and ML algorithms help to detect and analyse enormous amounts of data, and the solutions they offer are more robust, flexible, and scalable.

The main targets of AI and ML based algorithms for cyber security are malware detection, network intrusion detection, and phishing and spam detection. Some of the major adopters of AI and ML based cyber security solutions are Google, IBM, Juniper Networks, Apple, Amazon, and Balbix. More and more companies are joining this bandwagon.

To sum up, integrating AI and ML into your cyber security solutions today is not an option but a necessity if you want to counter the emerging complex security threats.

Acknowledgements

Dr Behara would like to thank Santosh Shinde of BTIS, Enterprise Architecture division of HCL Technologies Ltd for giving the required time and support in many ways when this article was being written as part of Architecture Practice efforts. 

 **By: Dr Gopala Krishna Behara and Raja Sekhar Amirapu**

Dr Gopala Krishna Behara is an enterprise architect in the BTIS Enterprise Architecture division of HCL Technologies Ltd. He has 'a total of' 27 years of experience in the IT industry.

Raja Sekhar Amirapu is a senior architect at Tech Mahindra. He has 'a total of' 26 years of experience in the IT industry.

Disclaimer: The views expressed in this article are that of the authors and HCL or Tech Mahindra do not subscribe to the substance, veracity or truthfulness of the said opinion.

THE COMPLETE MAGAZINE ON OPEN SOURCE

OpenSource
ForYou

The latest from the Open Source world is here.

OpenSourceForU.com

Join the community at facebook.com/opensourceforu

Follow us on Twitter @OpenSourceForU